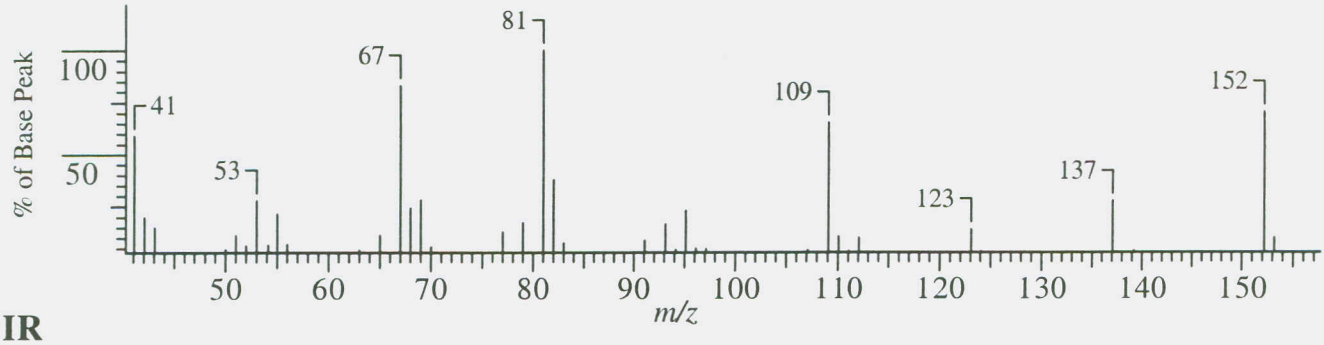
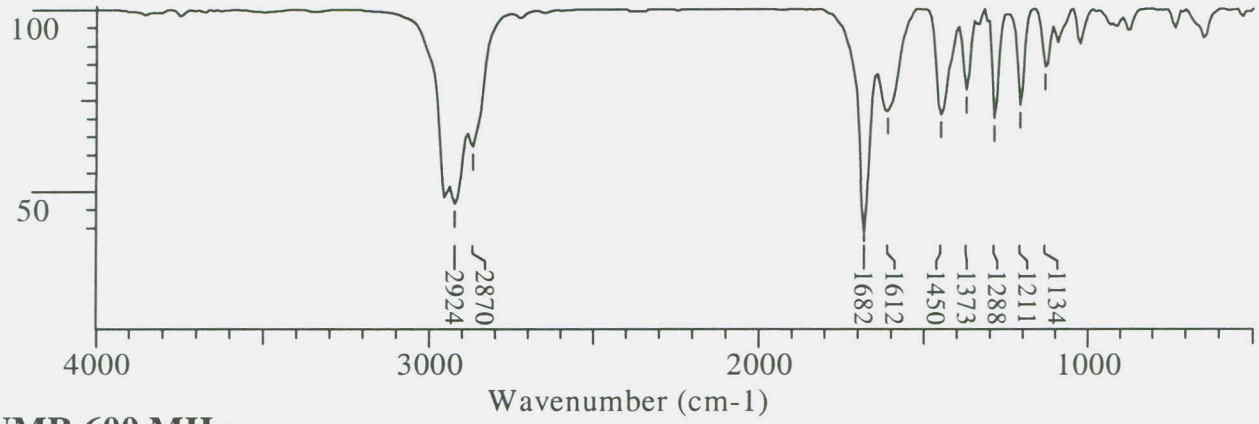
Organic Cume
February 6, 2009

The following pages contain spectra for an unknown compound. List as much structural information as you can, including the molecular formula, index of hydrogen deficiency, and structural fragments with justification. Using the spectra, identify spin patterns present and postulate a structure consistent with the data.

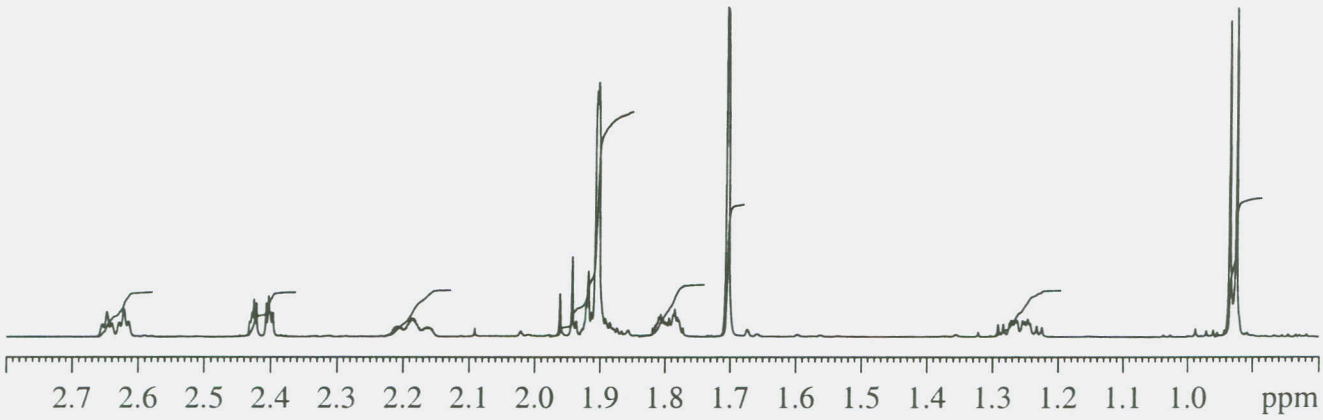
MASS



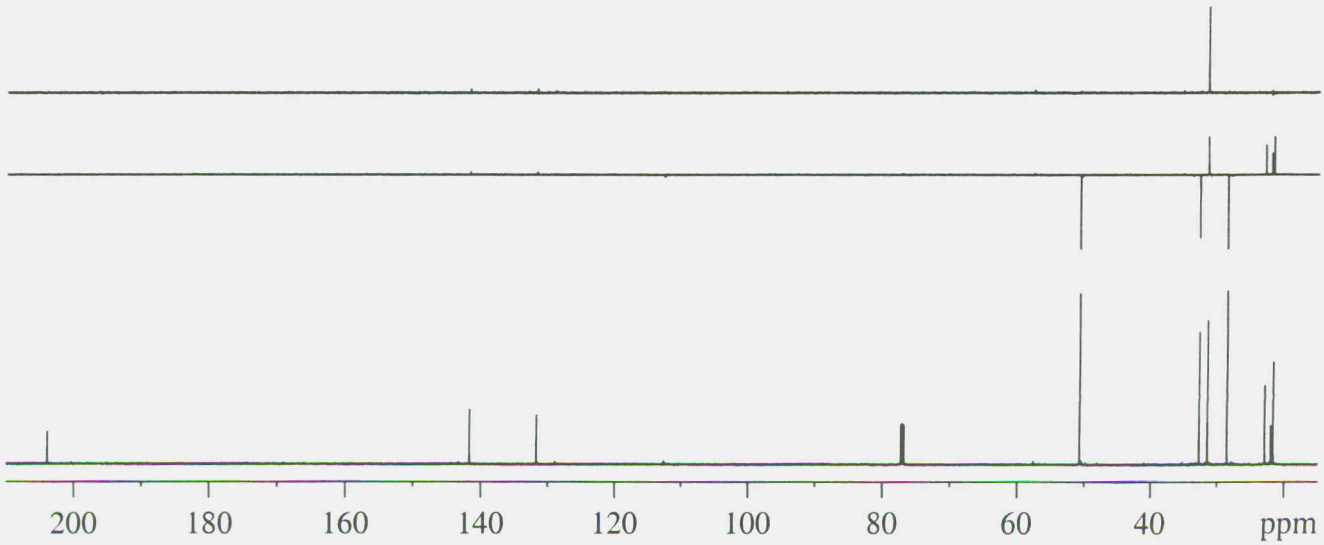
IR

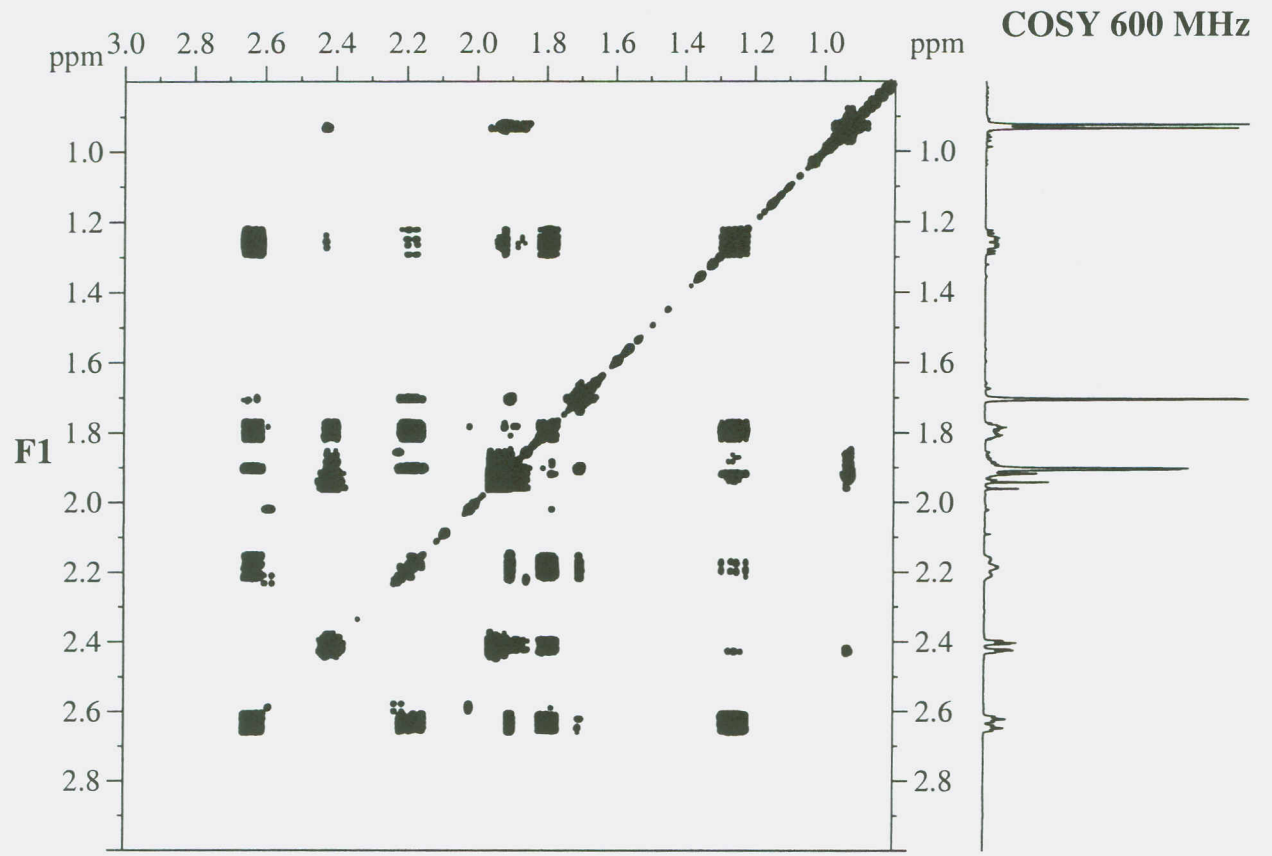


^{1}H NMR 600 MHz



^{13}C/DEPT NMR 150.9 MHz

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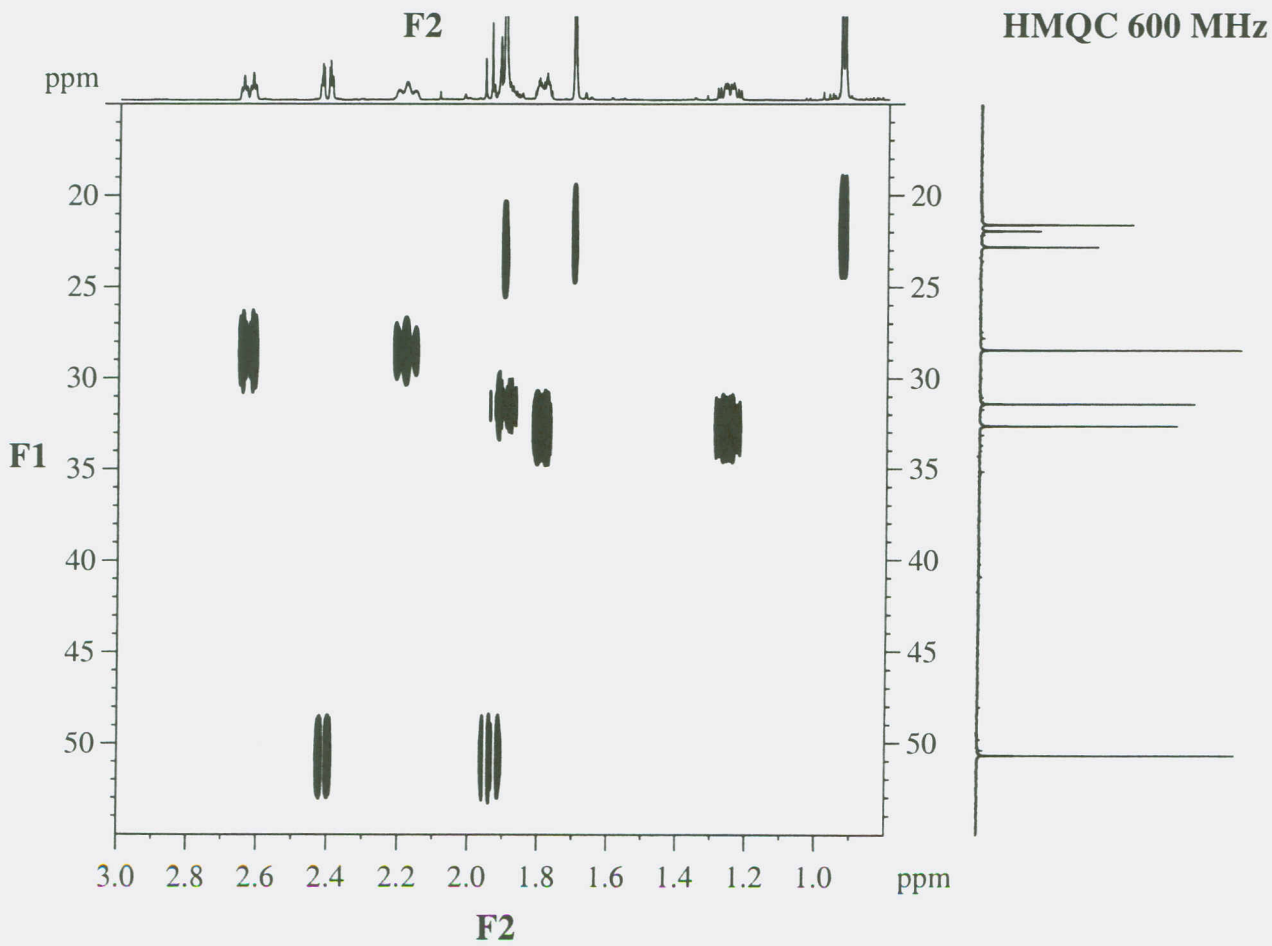
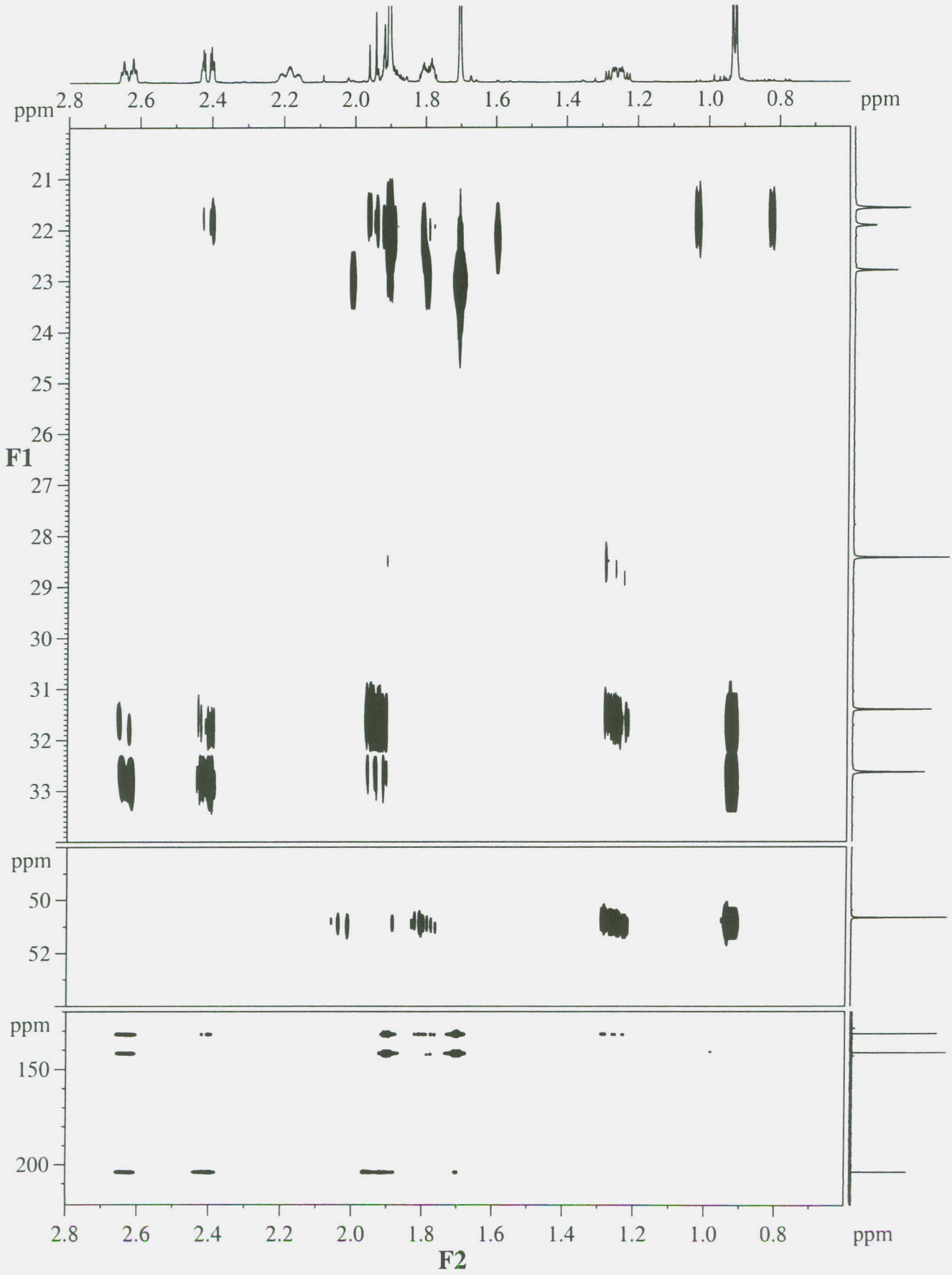
F1
F2


HMQC 600 MHz
ppm
20 25 30 35 40 45 50
F1
3.0 2.8 2.6 2.4 2.2 2.0 1.8 1.6 1.4 1.2 1.0
F2

D-3

HMBC 600 MHz



n-4